

Data Wrangling 🤠

Introduction to Data Science

Lesson 12

Quick Poll

Where does *data* show up in your own life? Think Spotify, sports, weather apps, college admissions, video games...

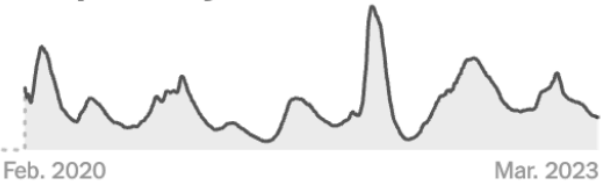
New reported cases

All time

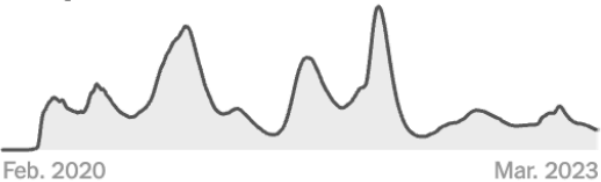
Last 90 days



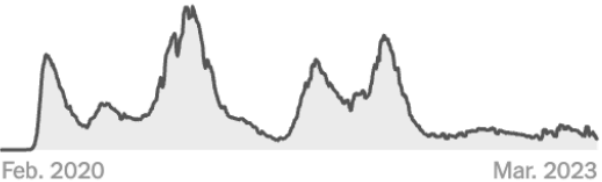
Test positivity rate



Hospitalized



Deaths



Durham

81°

Lower temperatures expected tomorrow, with a high of 87°.

Now	8:30PM	9PM	10PM	11PM	12AM
					
81°	Sunset	35%	25%	77°	76°

10-DAY FORECAST

Today	 50%	76° ————— 93°
Mon	 90%	74° ————— 87°
Tue	 75%	74° ————— 91°

What *is* Data Science?

- **Data science** is about drawing useful conclusions from data using computation.
- We'll use Python to *explore data* including data manipulation and visualization 
- Use data from the past to predict what may happen in the future (i.e. machine learning) 

What *makes up* Data Science?

Data science sits at the intersection of:

- **Programming** to collect, clean, and process data
- **Statistics** and **math** to find patterns in data
- **Domain expertise** to know what questions to ask!

Data Workflow

1. **Ask** a question
 2. **Collect** the data you need
 3. **Clean** / wrangle it into a usable shape
 4. **Explore** and analyze for patterns
 5. **Communicate** the result — a chart, a number, a decision
- Data scientists often say cleaning and wrangling data takes up **most** of a project's time — sometimes estimated at 60–80%.

Structured vs. Unstructured Data

- **Structured:** fits neatly into rows and columns
 - spreadsheets, CSVs, JSON, database tables
- **Unstructured:** doesn't have a fixed shape
 - text, images, audio, video

```
# A table of grades
students = [
    {"name": "Alicia", "grade": 91},
    {"name": "Vrinda", "grade": 84},
]
```


Tabular Data

- A subset of structured data
- **Individual** (row): Person/place/thing for which data is recorded
- **Variable** (column): Something that is recorded for each individual

State	Region	Capital City	Population	Land Area	Party	Density
Alabama	South	Montgomery	5024279	50645	Republican	99.21
Alaska	West	Juneau	733391	570641	Republican	1.29
Arizona	West	Phoenix	7151502	113594	Republican	62.96
Arkansas	South	Little Rock	3011524	52035	Republican	57.87
California	West	Sacramento	39538223	155779	Democratic	253.81
Colorado	West	Denver	5773714	103642	Democratic	55.71
Connecticut	Northeast	Hartford	3605944	4842	Democratic	744.72

individual

variable

Types of Variables

- There are two main types of variables:
 - **Numerical:** It makes sense to do arithmetic or math with the values
 - **Categorical:** Values fall into categories, that may or may not have some order to them

Name that Type!

- Salaries of NBA players 🏀
 - **Individual:** An NBA player
 - **Variable:** Their salary
- Movie genres 🎬
 - **Individual:** A movie
 - **Variable:** Its genre

Name that Type!

- Company's annual profit 💰
 - **Individual:** A company
 - **Variable:** Its annual profit
- Zip codes 🏠
 - **Individual:** US resident
 - **Variable:** Zip code

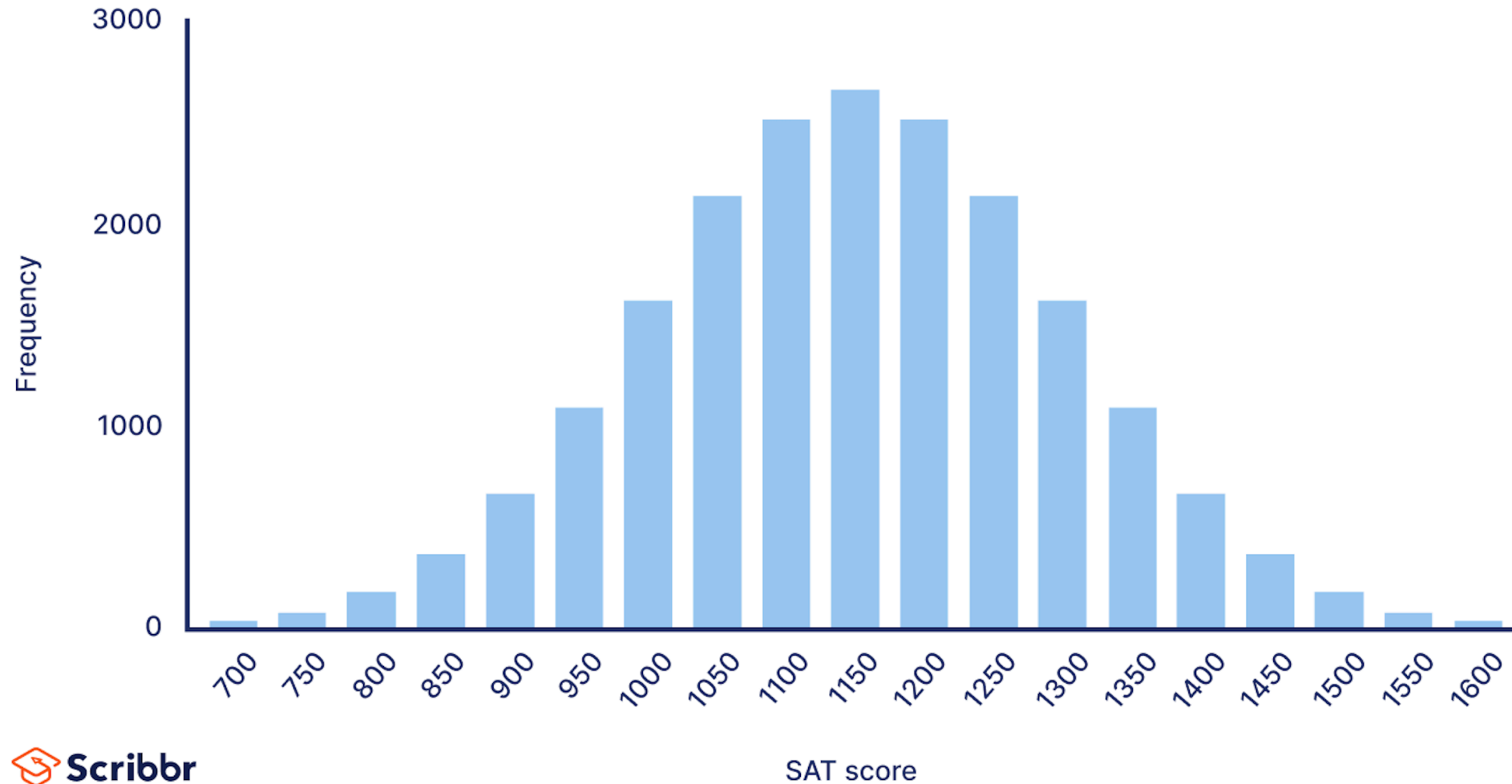
Name that Type!

- Flu shots administered per day 
 - **Individual:** Date
 - **Variable:** Number of flu shots administered on that date
- Level of prior programming experience for students 
 - **Individual:** A Summer Accelerator student
 - **Variable:** Their level (e.g. *none, low, medium, or high*)

Distribution of Variable

- The **distribution** of a variable consists of all values of the variable that occur in the data, along with their frequencies
- Distributions help you understand: *How often does a variable take on a certain value?*
- Both **categorical** and **numerical** variables have distributions

Example of normal distribution



Descriptive Statistics

- Descriptive statistics (*mean, median, mode, std dev, range, etc.*) are numeric summaries of a variable's distribution
- They describe a distribution's **center** and **spread** without having to plot it

Measures of Center

- **Mean:** Sum all values and divide by the total number of values
- **Median:** The middle value when all of the data is sorted
- **Mode:** The most frequently occurring value in the dataset

Concept Check

What is the **mean**, **median**, and **mode** of the following dataset

`[5, 10, 10, 15, 20]`?

Measures of Spread

- The *spread* describes how wide a distribution is
- **Range:** The difference of the largest and smallest value in the dataset
- **Variance:** The average squared distance from the mean
- **Standard Deviation:** Take the square root of the variance

Concept Check

What is the standard deviation of the following dataset

`[1, 2, 3, 4, 5]`?

- Find the mean.
- Subtract the mean from each value to get the *deviations*.
- Square each deviation.
- Average the squared deviations — that's the variance.
- Take the square root of the variance — that's the standard deviation.

Other Information

- **Percentile:** The value below which a given percentage of the data falls
- **Proportion:** A value between 0 and 1 which represents the fraction of the dataset that meets a certain condition

Concept Check

What is the 50th percentile of the following dataset

`[10, 20, 30, 40]`?

Inferential Statistics

- Making conclusions about a population based on data from a random sample
- Beyond the scope of this class!